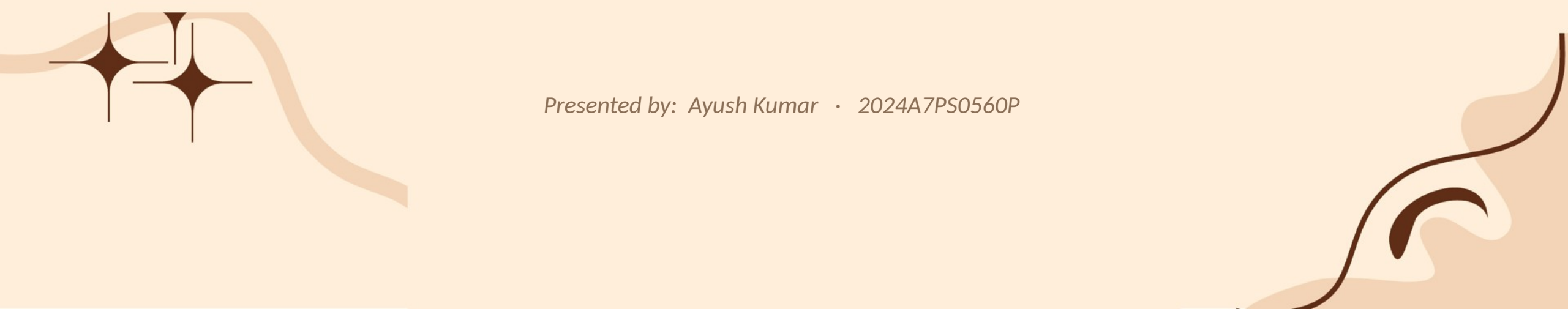




Robust Deepfake Detection Across Image Modalities

Practice School-I • C-DAC CINE, Guwahati



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THE PROBLEM

In-domain (StyleGAN)

>99%

ViT-Base/16 on 140K real/fake faces — ROC-AUC $\approx$ 1.0.

Unseen generators (DFFD)

~30%

Recall on fakes it never saw. Most fakes pass as real.

Why it happens

Generators are diverse — synthesis (StyleGAN, PGGAN), attribute edits (StarGAN, FaceApp) and diffusion each leave different traces.

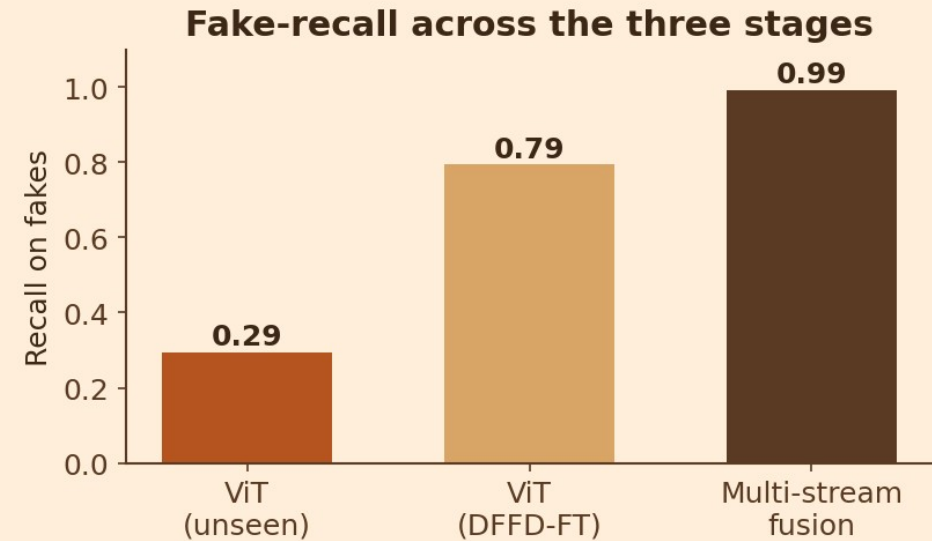
Detectors overfit one fingerprint — a model trained on one generator learns that generator, not “fakeness.”

The failure is silent — precision stays high while recall collapses, so the model quietly waves fakes through.

QUANTIFYING THE GAP

Metric	Before	After
Accuracy	0.646	0.894
Precision	0.997	0.994
Recall	0.294	0.793
F1	0.454	0.882
ROC-AUC	0.919	0.964

Recall on fakes falls to 0.29, then recovers to 0.79 after seeing diverse fakes. Precision barely moves — the model was under-detecting, not over-flagging.



Fake-recall across the three project stages.

THE HARD CASE: FACEAPP

Fake type	Before	After
FaceApp (edit)	0.380	0.452
StarGAN (edit)	0.252	0.760
PGGAN (synthesis)	0.252	0.960
StyleGAN (synth.)	0.292	1.000

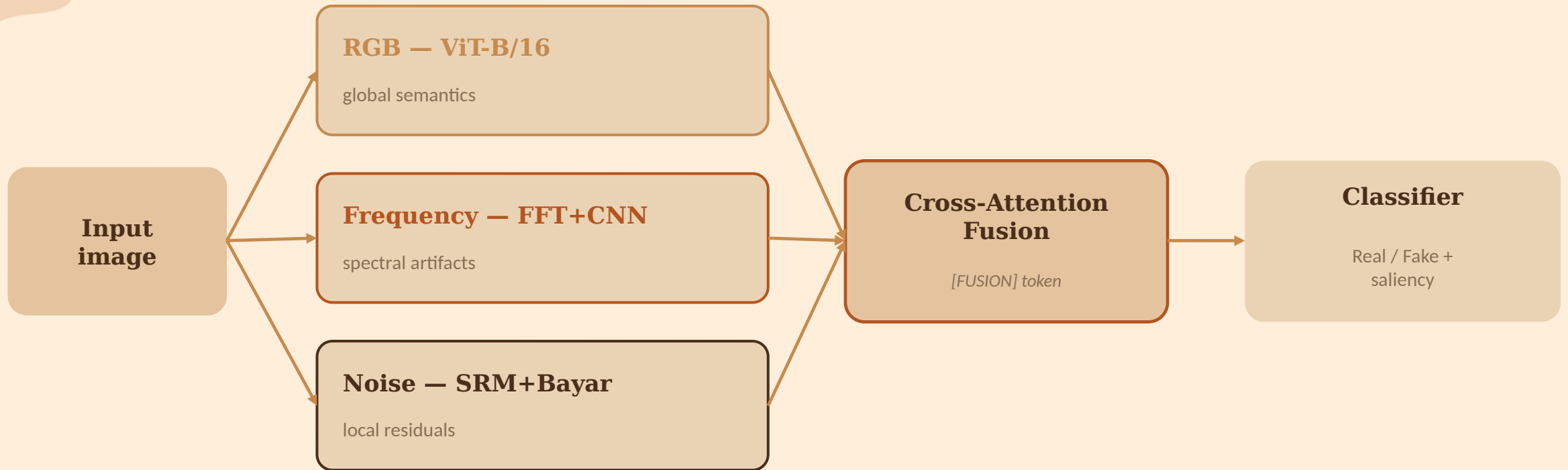
Why FaceApp resists

FaceApp doesn't generate a face — it **edits a real one**. A smile, an age, a hairline; the rest of the photo is untouched.

So there is no global inconsistency for a semantic model to catch — the image essentially *is* a real photo with one region that isn't.

This is what motivates the frequency and noise-residual streams.

THE ARCHITECTURE



Deep supervision: every stream keeps its own auxiliary head during training, so the strong pretrained RGB stream can't drown the two new ones.

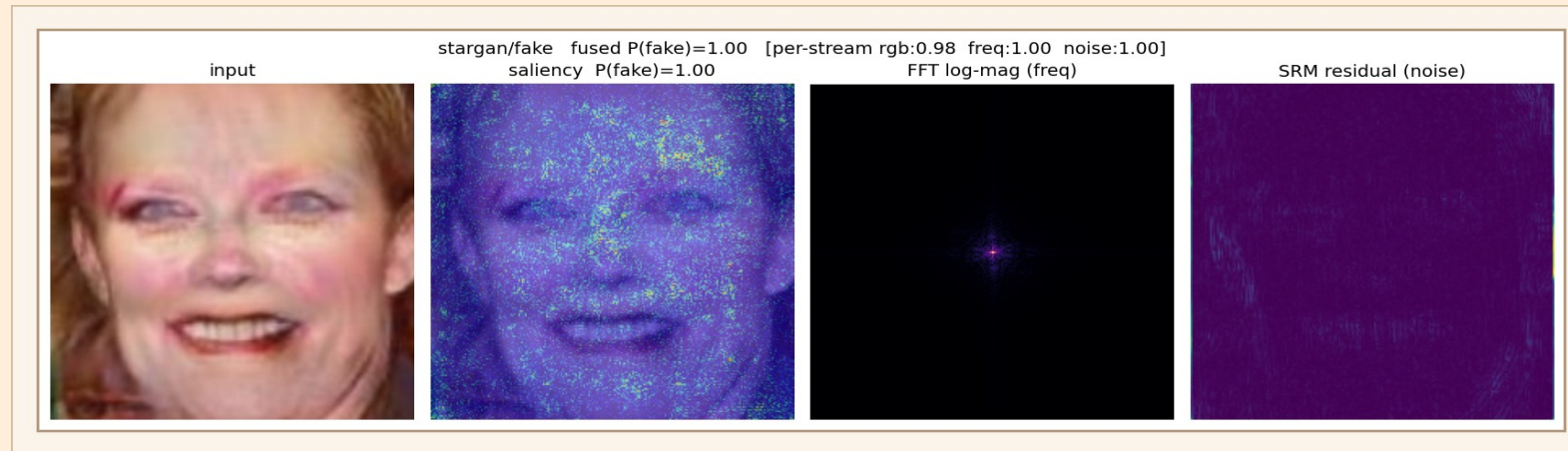
READING THE NON-SEMANTIC CUES

Frequency — the 2-D FFT log-magnitude spectrum. GAN upsampling leaves periodic grids invisible in RGB.

Noise — fixed SRM high-pass kernels + a learnable Bayar conv expose local noise / splice inconsistency.

```
fft = torch.fft.fft2(x, norm="ortho")
mag = torch.log1p(torch.abs(fft))
# fp32 always – fp16 FFT is unstable
```

```
k = [[ 0, 0, 0, 0, 0],
      [ 0, -1, 2, -1, 0],
      [ 0, 2, -4, 2, 0]] / 4.0
# one of 3 fixed SRM filters
```



A real DFFD example — the two right panels are exactly these streams' inputs.

CROSS-ATTENTION FUSION

```
def forward(self, stream_tokens):
    seq = [self.fusion_token.expand(B, -1, -1)]
    for name, tok in stream_tokens.items():
        t = self.proj[name](tok) \
            + self.stream_embed[name]
        seq.append(t)
        aux[name] = self.aux_heads[name](
            t.mean(1)) # deep sup
    x = torch.cat(seq, dim=1)
    x = self.encoder(x) # self-attn =
                        # cross-attn
    logits = self.head(self.norm(x[:,0]))
    return logits, aux
```

1. Stream-type embeddings

each token is tagged with its modality.

2. Concatenate + self-attend

every RGB token attends to every frequency / noise token
— this is the cross-attention.

3. Read the fusion token

its state feeds the classifier; aux heads keep each stream alive.

TRAINING METHODOLOGY

Stage 1

Warm-start

Train frequency and noise encoders alone, so they enter fusion already useful.

Stage 2

Freeze + fuse

Freeze the DFFD-tuned RGB ViT; train only fusion and the new streams.

Stage 3

(dropped)

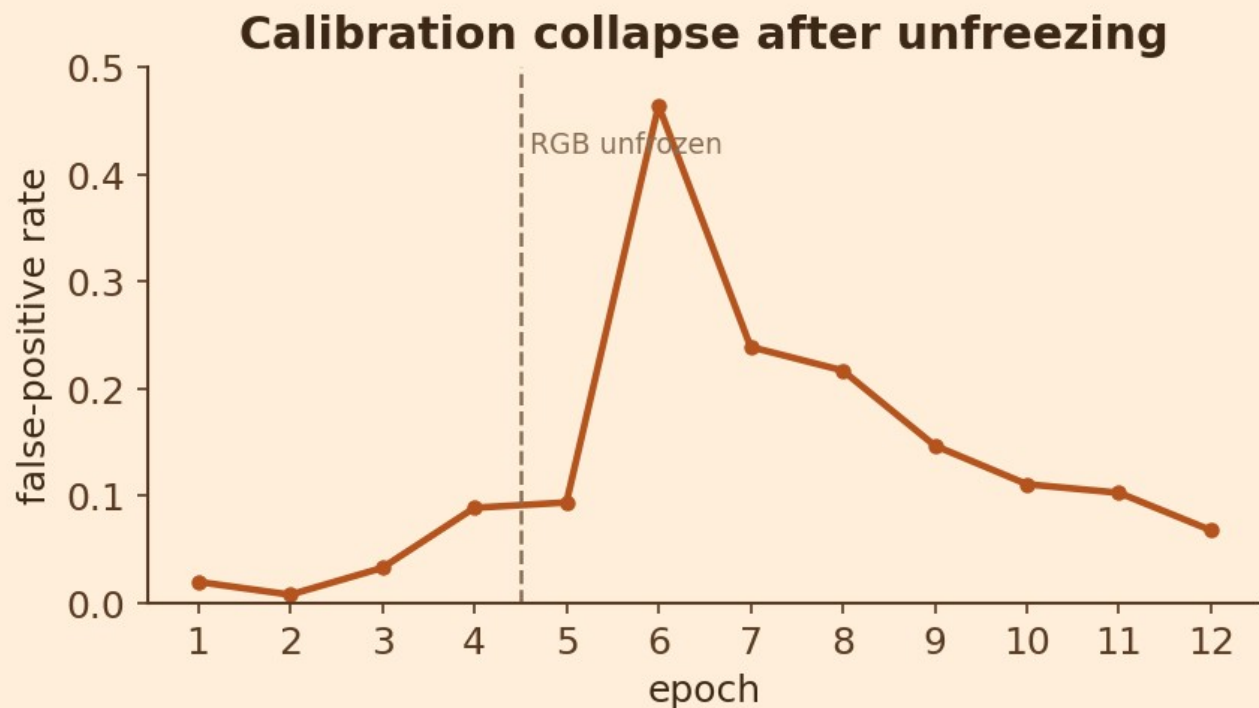
End-to-end unfreeze was tested — but it hurt. Final recipe keeps RGB frozen.

```
loss = CE(fusion, y) +  $\lambda$  · mean( CE(aux_k, y) for each stream k )
```

```
# AdamW · cosine LR · mixed precision (FFT/SRM forced to fp32)
```

```
# discriminative LR: RGB backbone at 0.1x base rate when unfrozen
```


A FINDING: FROZEN BEATS FINE-TUNED



The model peaked at epoch 1 while frozen. After unfreezing it flagged up to 46% of real faces as fake and never recovered.

Epoch 6

46%

real faces wrongly flagged fake

The takeaway

A confident, already-tuned backbone is destabilised by joint fine-tuning alongside fresh streams.

Final recipe: RGB frozen throughout — more accurate, and ~3x faster per epoch.

RESULTS

Held-out DFFD test set — 53,381 images, identity-disjoint split.

0.992

Recall

0.993

F1

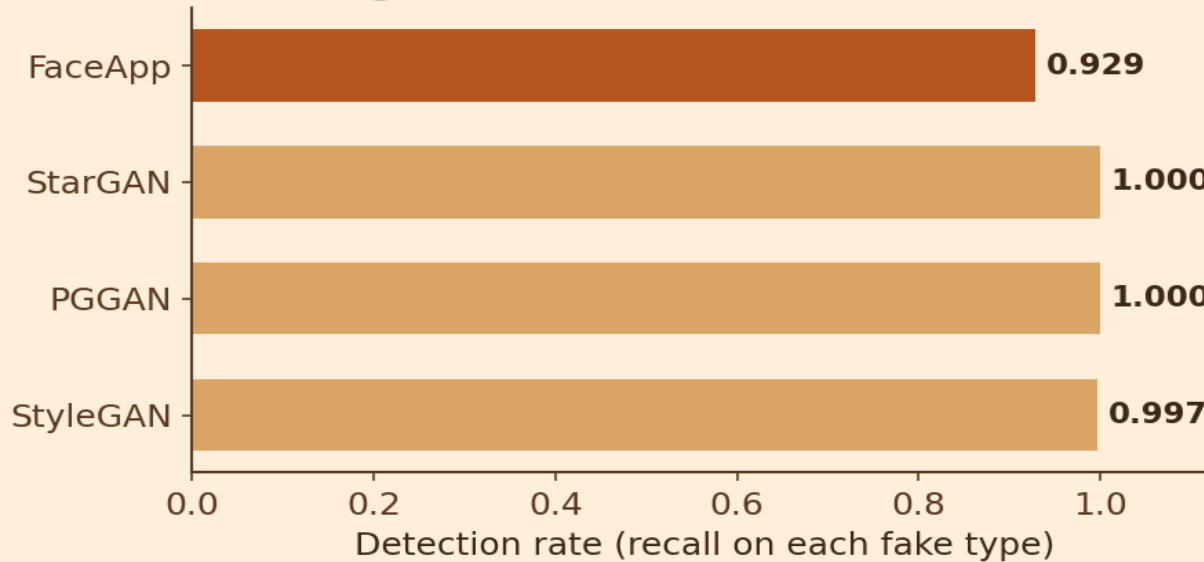
0.999

ROC-AUC

0.989

Accuracy

Per-generator detection — DFFD test



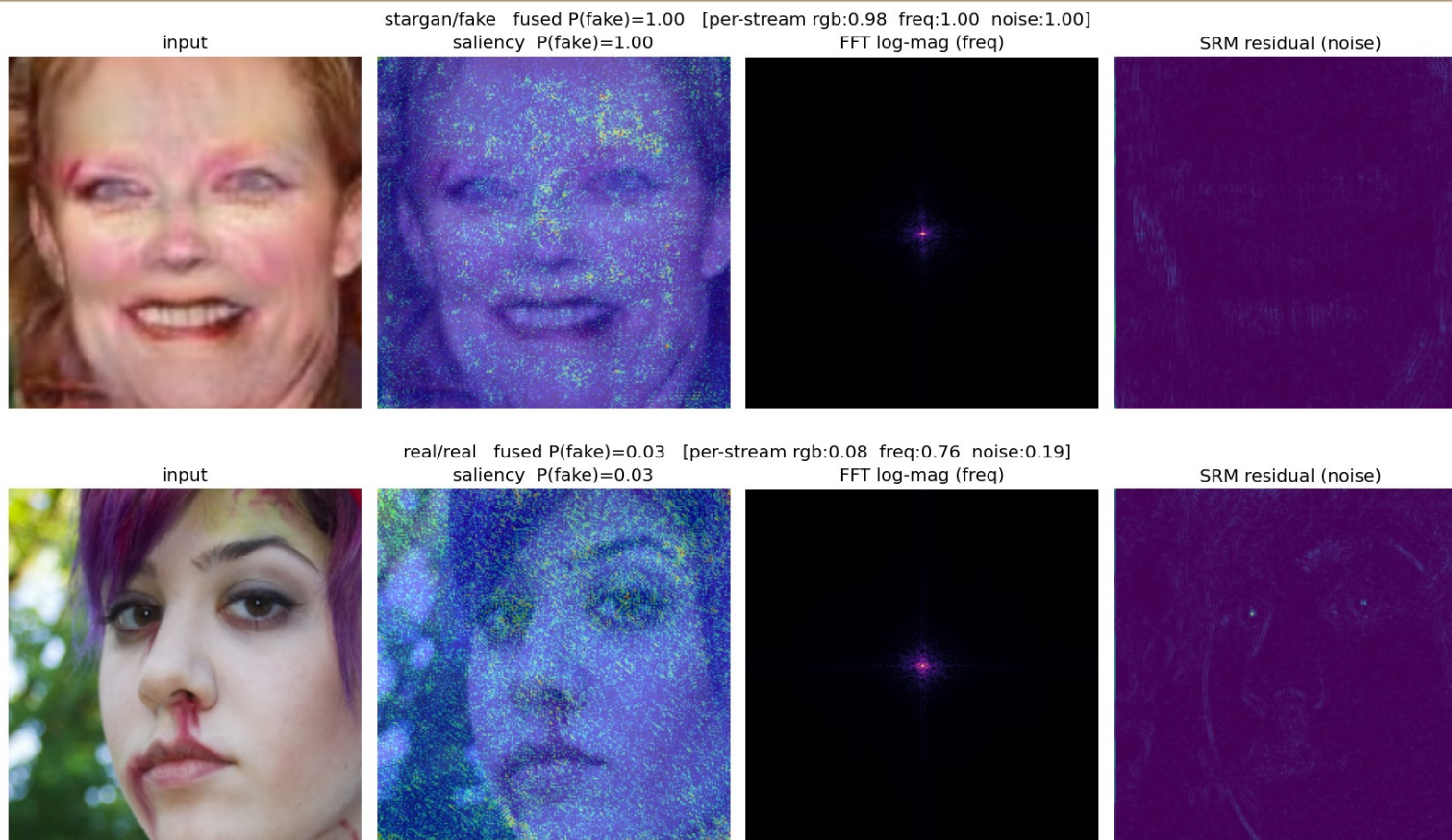
Read the FaceApp row

Synthesis generators are saturated (≈ 0.997 – 1.000).

FaceApp — the category that motivated the design — rises to **0.93**, up from the single-stream 0.45.

False positives on real faces: just 2.6%.

EXPLAINABILITY



input · saliency · FFT spectrum · SRM residual

Saliency

the pixels that moved P(fake).

Stream views

the FFT & SRM maps each stream reads.

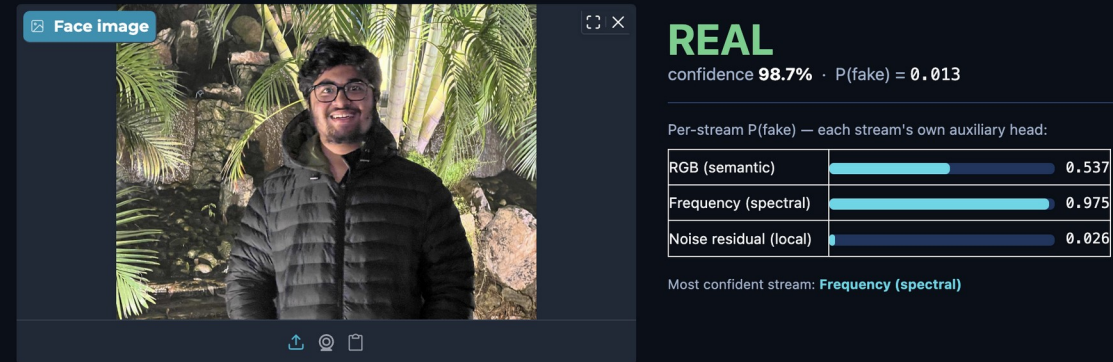
Per-stream score

which stream drove the call.

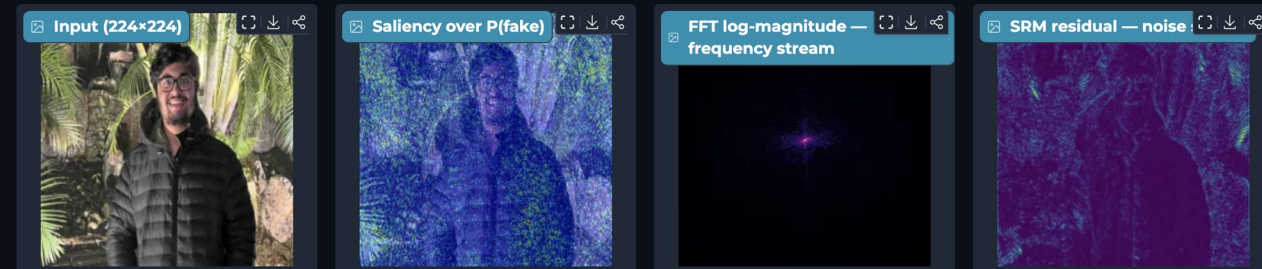
INTERACTIVE VERIFICATION TOOL

Multi-Stream Deepfake Detector

Upload a face image. The model reads it three ways — semantic (RGB ViT), spectral (FFT), and local noise residual (SRM) — and fuses them with cross-attention. The panels below show what each stream sees, so the verdict can be attributed rather than just asserted.



What the model looked at



A bright saliency region marks pixels that most changed the fake probability. The FFT panel exposes upsampling grids and GAN fingerprints; the SRM panel exposes local noise inconsistency, which is what

Live demo

Built with Gradio.

Inference mirrors training exactly — raw $[0,1]$ tensors at 224×224 .

This real example: **REAL** at 98.7% confidence, streams reasoning independently.

CONCLUSIONS

1 In-domain accuracy misleads

A ViT at >99% in-domain fell to ~30% recall on unseen generators.

2 Complementary evidence wins

Fusing spectral + noise views took fake recall to 0.99 on held-out test data.

3 Local edits are hardest

FaceApp — editing a real photo — stays weakest at 0.93 and drove the design.

4 Frozen beats fine-tuned

Unfreezing destabilised calibration; freezing was more accurate and faster.

Future work: diffusion-generated faces · video benchmarks (FaceForensics++, Celeb-DF) · continual learning for new generators.

DELIVERABLES

Trained models

DFFD-tuned ViT, warm-started frequency & noise streams, final fusion checkpoint.

Open codebase

Manifest builder, staged training, evaluation, explainability — one documented repo.

Explainability tool

Saliency + FFT/SRM views + per-stream attribution for any image.

Live interface

A Gradio app for image verification, mirroring the training pipeline.

The slide features a light beige background with decorative elements in the corners. The top-left and bottom-right corners have abstract, flowing line art in a dark brown color. The top-right corner features a branch with several leaves, also in dark brown. The bottom-left corner has two stylized starburst or spark motifs. The text is centered on the slide.

THANK YOU

Questions & discussion

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